

ADVOCATE

Autonomous Multi-Agent SDLC System

Specialized AI agents collaborating autonomously to Plan, Code, Critique, Test, and Release production-ready code.

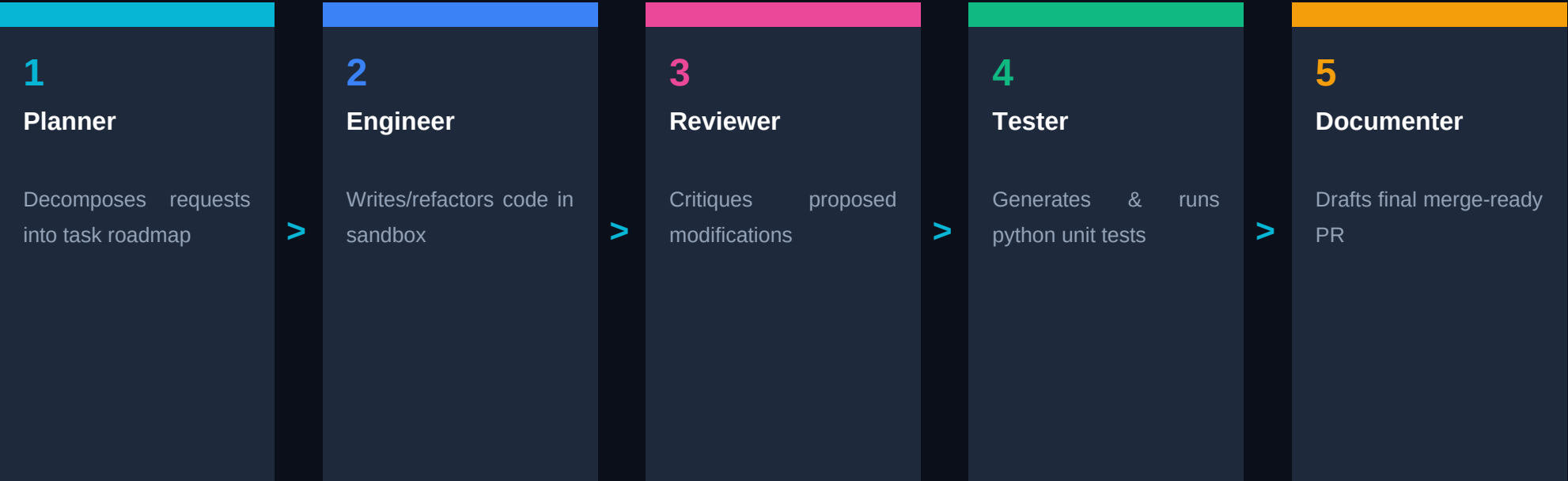
The Core Problem & Solution

Traditional Bottlenecks

- + Slow manual Cross-Review leading to QA delays.
- + Tested and changes slip into prod release.
- + Lack of immediate visibility into development pipeline.
- + High cognitive load mapping requirements to steps and manually.

ADVOCATE Solution

Multi-Agent Collaboration Architecture



Shared Event Message Bus

Coordinates communication, maintains system state, and routes task handoffs between agents. Pushes live updates directly to the web dashboard via real-time WebSockets.

Key Advanced Capabilities

Cross-Model Review loops

Runs reviewer agents on secondary LLM engines (e.g. Claude Haiku reviewing Claude Sonnet's output) to minimize same-model cognitive bias.

Secure Subprocess Sandboxing

Executes unit test suites inside isolated workspace folders with path-traversal protection. Fallback handles lack of global 'uv' natively.

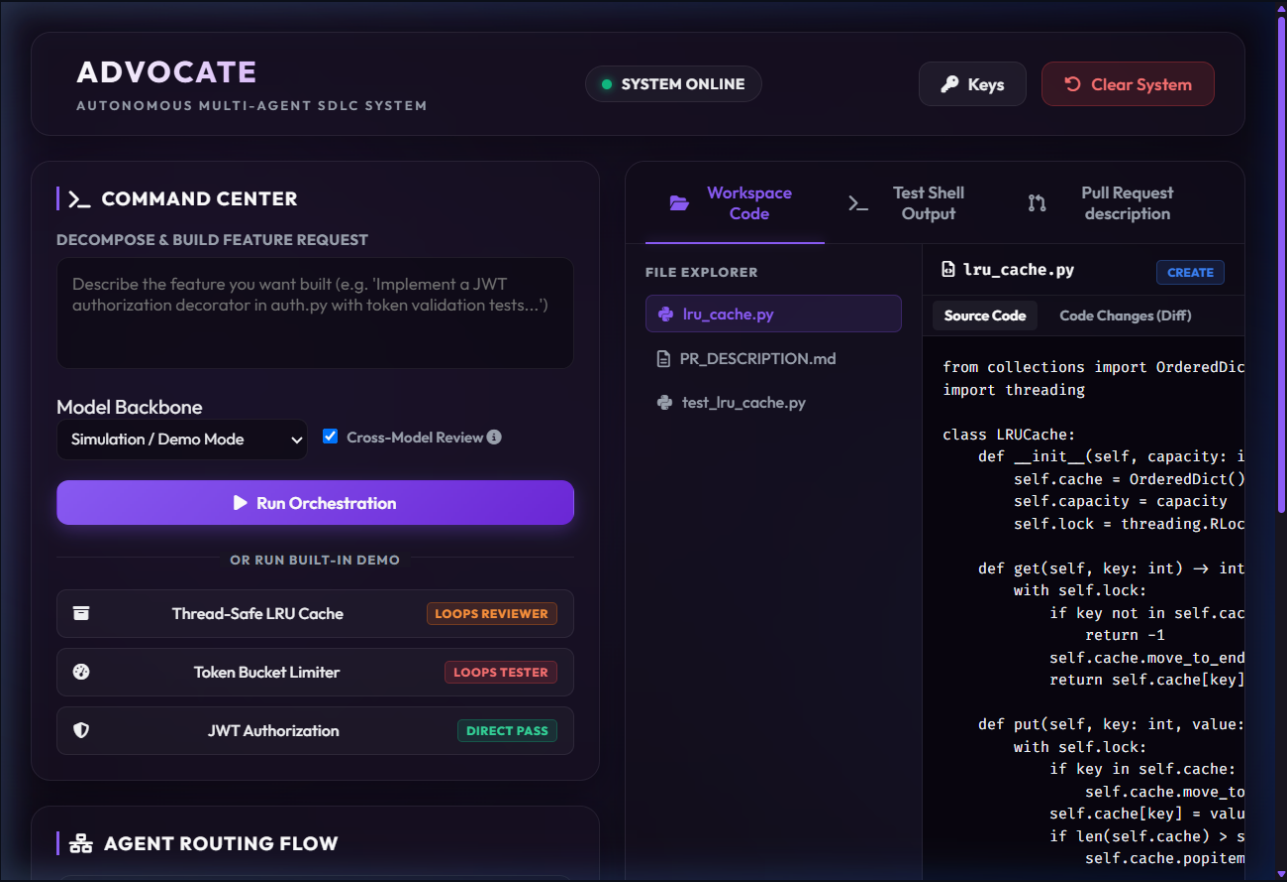
Live Interactive Dashboard

Glassmorphic interface offering live event timeline logs, file structure tree explorers, inline visual code diffs, and test terminal output.

Event-Driven Architecture

FastAPI + WebSocket message broker allows developers to observe autonomous handoffs, test logs, and PR releases in real time.

Glassmorphic Web Dashboard



Control Center Elements

- Command Center:
Launch demo scenarios or submit prompt features
- Agent Routing Flow:
Visually displays active routing steps
- Workspace File Explorer:
Explore generated files and read code
- Code Diff Viewer:
Read line-by-line colored diff changes
- Test Terminal Output:
Review sandboxed runtime test logs
- Pull Request Tab:
Displays the autogenerated markdown description

Deployment & Production Configuration

Render Cloud Deploy

Uses `FormData` and `FileList` keys to pass a prepared `transferable` box, `localStorage` or injected as `Secure Beaker Build` and `Variables`.

- Built up in a modular way to meet requirements.
- Securely store sensitive information dynamically.
- Securely store sensitive information in a safe way.
- Secure HTTPS auto-redirect support.

Docker Container

Key Security Features